

Andrew Joros

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SUMMARY

Research computing engineer who builds production-grade data and analytics systems from ingestion to decision-ready delivery. I develop reproducible pipelines, APIs, and interactive applications that turn complex datasets into reliable, actionable insights for technical and non-technical stakeholders.

EXPERIENCE

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|---|----------------|
| Research Computing Engineer (Software Development / Data Scientist) | 2024 – Current |
| <i>Division of Atmospheric Science / Division of Hydrologic Sciences, Desert Research Institute</i> <i>Reno, NV</i> | |
| <ul style="list-style-type: none">• Contributed to predictive analytics by validating clustering outputs and automating preprocessing• Built OpenFOAM post-processing pipeline for snow-transport datasets and wall-shear diagnostics• Operated automated snow-drought pipeline producing HUC and station classification graphics• Developed internal DRI-hosted Flask/xarray web app for county climate-risk maps• Automated Nevada groundwater and projected pumpage reporting across 262 hydrographic basins• Built public snowpack-runoff decision tool using Leaflet/Plotly and SNOTEL ingestion | |
| Research Computing Engineer (Cloud Computing / Data Scientist) | 2019 – 2023 |
| <i>Center for Genomic Medicine & Climate Ecosystem Fire Applications, Desert Research Institute</i> <i>Reno, NV</i> | |
| <ul style="list-style-type: none">• Provided cloud computing and data engineering support for genetic analysis projects• Worked with team members on cloud-based data engineering and statistical projects, improving productivity• Led AWS onboarding and training, enhancing the team's cloud computing skills• Created AWS-based solutions for statistical smoke dispersion models for California's fire management• Developed a dynamic front-end webpage using AWS, presenting statistical modeling outputs related to Interagency Wildland Fire Air Quality, empowering stakeholders with valuable insights into air quality conditions | |
| Staff Research Scientist (Software Development / Data Scientist) | 2016 – 2019 |
| <i>Applied Innovation Center, Desert Research Institute</i> <i>Reno, NV</i> | |
| <ul style="list-style-type: none">• Developed groundwater survey software using vertical electrical sounding to improve well siting and drilling• Contributed to meteorological and air pollutant data pipelines supporting Healthy Nevada Project research• Integrated Christopher Ranch's sensor data into AWS for storage and machine learning use• Leveraged AWS EC2 to devise and implement cutting-edge agricultural insect and disease probabilistic models, empowering customers with accurate predictions for effective pest management and crop protection• Created an automated AWS sensor alert system for data integrity and reliability | |
| Assistant Research Scientist (High Performance Computing) | 2015 – 2016 |
| <i>Applied Innovation Center, Desert Research Institute</i> <i>Reno, NV</i> | |
| <ul style="list-style-type: none">• Constructed backend data architecture for WINDS, improving weather processing and decision support• Developed a web portal for NDOT, enabling traffic-data analysis for transportation planning• Led NevCAN's data visualization project, presenting real-time climate and ecological data effectively• Created an image processing algorithm for greenhouse experiments to analyze vegetation health and growth | |
| Staff Scientist Programmer, Hydrometeorologist | 2013 – 2015 |
| <i>Desert Research Institute</i> <i>Reno, NV</i> | |
| <ul style="list-style-type: none">• Used Earth Engine JavaScript API for geospatial data analysis in hydrometeorology• Contributed to Climate and Integrated Earth Monitoring Engine dashboard, improving data visualization• Processed Landsat and weather data via cloud computing, aiding Nevada's ecosystem analysis and conservation• Evaluated precipitation datasets against Texas weather stations to improve irrigation and water management | |
| Research Assistant | 2009 – 2013 |
| <i>Desert Research Institute</i> <i>Reno, NV</i> | |
| <ul style="list-style-type: none">• Investigated monsoonal surges in the Northern Great Basin using statistical and dynamic methods• Improved weather datasets to better represent irrigated agricultural environments• Worked with CEFA to create a RAWs network density plot, aiding land managers' decisions• Developed web frameworks for NOAA-funded drought monitoring and an El Nino climate risk project | |
| Statistical Analyst | Summer 2009 |
| <i>Atmospheric Systems Corporation</i> <i>Valencia, CA</i> | |
| <ul style="list-style-type: none">• Conducted statistical QA/QC of SODAR output data to ensure reliability | |

EDUCATION

University of Nevada

M.S. in Atmospheric Science

Reno, NV

2009 – 2011

San Jose State University

B.S. in Meteorology

San Jose, CA

2006 – 2009

SKILLS

Programming Languages: Python, MATLAB, R, JavaScript, SQL

Data & Scientific Computing: xarray, pandas, NumPy, NetCDF, Parquet, JSON

Modeling & Visualization: OpenFOAM, ParaView, PyVista, Matplotlib, Plotly, ArcGIS

Web & APIs: Flask, Django, Leaflet, HTML5, CSS, REST APIs

Cloud & DevOps: Amazon Web Services, Google Cloud Platform, Azure, Git, GitHub Actions

Operating Systems: Linux (Ubuntu/CentOS/Redhat), Windows

PUBLICATIONS AND REPORTS

1. Yu, G., Miller, J. J., Mejia, J. F., Rasouli, K., McCarthy, M. I., Joros, A. N., O'Shea, K. (2026). Updating the Probable Maximum Precipitation for Nevada. Damon, N. E. (ed.), Division of Hydrologic Sciences, Desert Research Institute: Nevada Division of Water Resources, June 30, 2025, DHS publication #41312.
2. Luo, S., Jiang, R., Grzymski, J.J., Lee, W., & Joros, A. (2021). Comprehensive allele genotyping in critical pharmacogenes reduces residual clinical risk in diverse populations. *Clinical Pharmacology & Therapeutics*, 110(4), 899–911.
3. Kiser, D., Metcalf, W.J., Elhanan, G., Schnieder, B., Schlauch, K., Joros, A., Petersen, C. and Grzymski, J., 2020. Particulate matter and emergency visits for asthma: a time-series study of their association in the presence and absence of wildfire smoke in Reno, Nevada, 2013–2018. *Environmental Health*, 19(1), pp.1-12.
4. Nauslar, N.J., Hatchett, B.J., Brown, T.J., & Joros, A. (2019). Impact of the North American monsoon on wildfire activity in the southwest United States. *International Journal of Climatology*, 39(4), 1861–1879.
5. Abatzoglou, J.T., McEvoy, D.J., Joros, A., et al. (2017). The West Wide Drought Tracker: Drought Monitoring at Fine Spatial Scales. *Bulletin of the American Meteorological Society*, 98(9), 1815–1820.
6. Caldwell, T. G., Huntington, J. L., Scanlon, B., Joros, A. N., Howard, T., (2017): Improving Irrigation Water Use Estimates with Remote Sensing Technologies: An Initial Feasibility Study for Texas, 137 p., University of Texas, Bureau of Economic Geology: Austin, TX, Report prepared for Texas Water Development Board.
7. Joros, A., J. Abatzoglou, N. Nauslar, B. Hatchett, M. Kaplan, (2017) Extratropical Control of Monsoonal Surges in the Great Basin. *Monthly Weather Review* (In Progress)
8. Huntington, J. L., Gangopadhyay, S., Spears, M., Allen, R., King, D. L., Morton, C. G., Harrison, A., McEvoy, D. J., Joros, A. N., (2015): West-Wide Climate Risk Assessments: Irrigation Demand and Reservoir Evaporation Projections, Technical memorandum No. 68-68210-2014-01, 218 p, 754 appx, U.S. Bureau of Reclamation
9. Morton, C. G., Huntington, J. L., Allen, R. G., Kilic, A., Joros, A. N. (2015). More Landsat Satellites Equates to More Reliable Monitoring of Water Consumption. *Remote Sensing*.
10. Brown, T., Larkin, N.K., Rorig, M., & Joros, A. (2013). Improving Meteorological Data and Forecasts for Prescribed Fire Burn Day Decisions for the Lake Tahoe Basin. CEFA Final Report (SNPLMA).
11. Joros, A., M. Kaplan, J. Abatzoglou (2011) Extratropical Control of Monsoonal Surges in the Great Basin. (Master's Thesis)

CONFERENCE PAPERS/PRESENTATIONS

1. Saftner, D. M., Arienzo, M. M., Atkin, J., Daniel, A. T., Dunkerly, C. W., Gordon, B., Greene, S. J., Huntington, J. L., Joros, A. N., Kozloski, R., Lutz, A. D., Parker, B. A., ReVelle, P., VanderMolen, K.A. (2026). Developing Water Resource Plans for Rural Nevada. Nevada Tribal Food and Agriculture Summit: Sparks, NV, March 24–27, 2026.
2. McEvoy, D. J., Rhoades, A., Huning, L., Heggli, A. E., Hatchett, B., Joros, A. N. (2025). Storm-scale impacts on seasonal snow droughts: A case study in the Pacific Northwest, United States. American Geophysical Union Annual Meeting: New Orleans, LA, December 17, 2025.

3. McEvoy, D. J., Rhoades, A., Huning, L., Heggli, A. E., Hatchett, B., Joros, A. N. (2025). Storm-scale impacts on seasonal snow droughts: A case study in the Pacific Northwest, United States. International Mountain Conference poster: Innsbruck, Austria, September 14–18, 2025.
4. Yu, G., Miller, J. J., Mejia, J. F., Rasouli, K., McCarthy, M. I., O'Shea, K., Joros, A. N. (2025). Updating the Probable Maximum Precipitation for Nevada. 2025 NWRA Annual Conference: Sparks, NV, January 27–30, 2025.
5. Hatchett, B., Abatzoglou, J., Joros, A. (2019). Monsoon-driven variability in fire–climate linkages. Presented at Western Governors' Association Fire & Climate Science meeting.
6. Rowan, C., Metcalf, W. J., Elhanan, G., Joros, A. N., & Grzymiski, J. J. (2018). Short-Term Sentinel Air Events in Relation to Health Care Utilization for Specific Health Conditions in Reno, Nevada. In B24. AN UPDATE ON INDOOR AND OUTDOOR AIR POLLUTION (pp. A2815-A2815). American Thoracic Society.
7. Rosenquist, N. A., Metcalf, J., Elhanan, G., Grzymiski, J., Joros, A. N., Field-Ridley, A., & Darrow, L. A. (2018, August). Acute Associations between PM_{2.5} and Allergic and Nonallergic Asthma Exacerbations in Children and Adults. In ISEE Conference Abstracts (Vol. 2018, No. 1).
8. Hegewisch, K., Daudert, B., Morton, C. G., Peterson, A., Joros, A. N., McEvoy, D. J., Erickson, T., Huntington, J. L., Abatzoglou, J. T. (2015). Google Drought Monitoring Through Cloud Computing and Visualization of Remote Sensing and Meteorological Datasets: Examples for California, American Geophysical Union Chapman Conference: Irvine, CA, April 20, 2015–April 22, 2015
9. Huntington, J. L., Daudert, B., Morton, C. G., McEvoy, D. J., Joros, A. N., Abatzoglou, J. T., Hegewisch, K., Peterson, A., Vansant, D. S., Allen, R., Kilic, A., Hobbins, M., Verdin, J. (2015). Cloud Computing for Drought Monitoring with Google Earth Engine, U.S. Drought Monitor Forum: Desert Research Institute, Reno, NV, April 14, 2015–April 16, 2015
10. Huntington, J. L., Morton, C. G., McGwire, K. C., Allen, R., Gorelick, N., Thau, D., Joros, A. N. (2014). Monitoring Groundwater Dependent Ecosystems in the Great Basin from Space and Clouds, American Water Resources Association Summer Specialty Conference: Reno, NV, May 31, 2014–July 2, 2014
11. Huntington, J. L., Morton, C. G., McGwire, K. C., Joros, A. N., Peterson, S., Gorelick, N., Thau, D., Allen, R. (2014). Cloud Computing of Landsat Imagery and Gridded Weather Data for Evaluating Groundwater Dependent Ecosystems in Nevada, Nevada Water Resources Association Annual Conference: Las Vegas, NV, February 2, 2014–February 2, 2014
12. Richard G. Allen, University of Idaho, Kimberly, ID; and J. L. Huntington, A. Kilic, H. Debruin, and A. Joros, (2013): Conditioning of NLDAS, NARR and arid weather station data to improve their representation of well-watered (reference) environments associated with irrigated agriculture. 93rd AMS Annual Meeting, 27th Conference on Hydrology, Austin, TX, 5–10 January 2013
13. Kelly T. Redmond, DRI, Reno, NV; and J.T. Abatzoglou, D. McEvoy, A. Joros, D. VanSant, and L. M. Edwards, (2013): The WestWide Drought Tracker: Drought Monitoring at Fine Spatial Scales 93rd AMS Annual Meeting, Austin, TX, 5–10 January 2013
14. Nauslar, Nick, J. Abatzoglou, A. Joros (2011): Development of an objective methodology for diagnosing Santa Ana winds. Ninth Symposium on Fire and Forest Meteorology, Palm Springs, CA, 18–20 October 2011
15. Joros, A., J. F. Mejia (2011): Impact of Eastern and Central Pacific ENSO events over the Continental United States. AMS Applied Climatology Conference 2011, Asheville, North Carolina, 18–20 July 2011
16. King, K. C., M. L. Kaplan, A. Joros, M. Liddle and E. Uher, (2011): Evaluation of the Operational Multi-scale Environment model with Grid Adaptivity (OMEGA) for use in Wind Energy Potential Assessment in the Great Basin of Nevada. Weather, Climate, and the New Energy Economy - Observations, Modeling, and Data for the Energy Sector. 91st AMS Annual Meeting, Seattle WA, 23–27 January 2010.
17. Abatzoglou, J.T., K.T. Redmond, L.M. Edwards and A. Joros (2011): Monitoring of drought in the western United States: metrics suitable for tracking drought historically and throughout the 21st century. 91st AMS Annual Meeting, Seattle WA, 23–27 January 2010.
18. Joros, A., J. Abatzoglou, J. Favors, B. Tan, M. Kaplan (2010): Extratropical Control of Monsoonal Surges into the Great Basin. AMS Mountain Meteorology Conference, Lake Tahoe, CA, 30 August – 3 September 2010.
19. Tan, B., A. Joros, and M. L. Kaplan, (2010): The Role of Midlatitude Circulations in Triggering Extratropical Convection during the 2004 North American Monsoon. 90th AMS Annual Meeting, Atlanta, GA, 17–22 January 2010.

20. King, K.C., M. Kaplan, A. Joros, M. Liddle, E. Uher (2010): Evaluation of the Operation Multiscale Model with Grid Adaptivity (OMEGA) for use in Wind Energy Potential Assessment in the Great Basin of Nevada. AMS Mountain Meteorology Conference, Lake Tahoe, CA, 30 August – 3 September 2010.
21. Uher, Erich, M. Kaplan, A. Joros, D. Decker (2010) Air Pollution Dispersion Forecasting: A Climatological Study of Cape Canaveral Tropospheric Wind Patterns. 90th AMS Annual Meeting, Atlanta, GA, 17-22 January 2010.